Aldehydes, Ketones and Carboxylic Acids

Carbonyl compounds matter as much as anything in organic chemistry: fabrics, flavourings, plastics and drugs are all built with them.

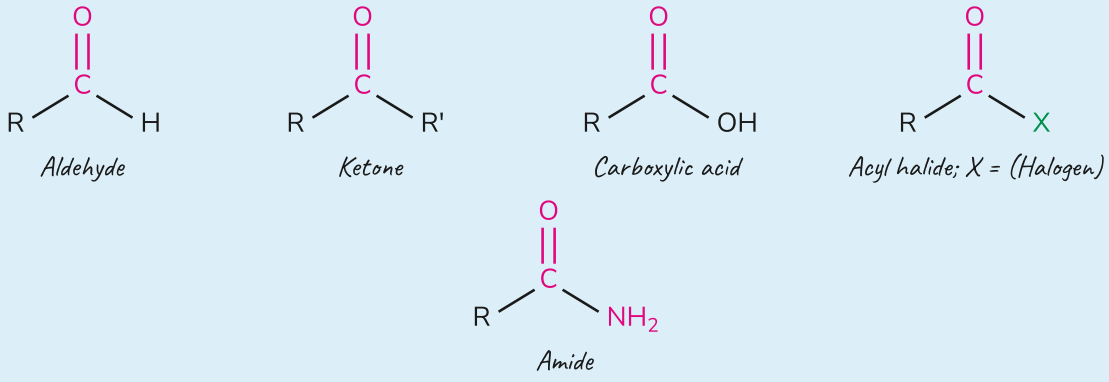
By the end of this unit

- write the common and the IUPAC name of an aldehyde, a ketone or a carboxylic acid
- draw the structures of compounds holding the **carbonyl** and the **carboxyl** functional group
- set out the chief methods of preparing these classes, and the reactions they give
- tie the physical properties and the chemical reactions of all three to their structures
- work through the mechanism of a few chosen reactions of aldehydes and ketones
- weigh what settles how acidic a carboxylic acid is, and what reactions follow
- say what aldehydes, ketones and carboxylic acids are used for

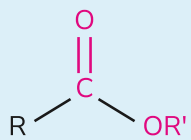
Where this unit starts from

- the previous unit dealt with functional groups holding a **carbon-oxygen single bond**
- this one takes the **carbon-oxygen double bond (>C=O)**, the **carbonyl group**, which counts among organic chemistry's most important functional groups
- in an **aldehyde** the carbonyl carbon is bonded to **a carbon and a hydrogen**; in a **ketone** to **two carbon atoms**
- where the carbonyl carbon holds **carbon or hydrogen on one side and the oxygen of an –OH on the other**, the compound is a **carboxylic acid**
- where it holds **carbon or hydrogen and the nitrogen of an $–NH_2$**, the compound is an **amide**; where it holds a **halogen**, an **acyl halide**
- **esters and anhydrides** derive from the carboxylic acids

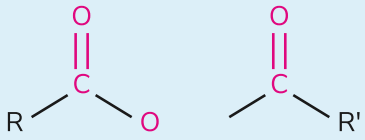
The general formulae



The unit runs in two parts: **I · Aldehydes and ketones**, then **II · Carboxylic acids**.

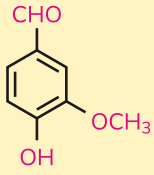


Ester

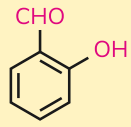


Acid anhydride

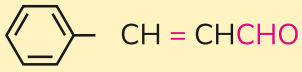
- aldehydes, ketones and carboxylic acids are **found throughout the plant and animal kingdoms**, and they play an **important part in the biochemical processes of life**
- they lend **fragrance and flavour** to nature: the vanilla bean gives **vanillin**, meadow sweet **salicylaldehyde**, cinnamon **cinnamaldehyde**, and all three smell pleasant



Vanillin



Salicylaldehyde



Cinnamaldehyde

- they go into **many food products and pharmaceuticals** to add flavour
- some members are **manufactured as solvents** — acetone for one — and for preparing **adhesives, paints, resins, perfumes, plastics and fabrics**

A.1 Nomenclature and structure of the carbonyl group

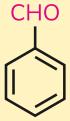
I · Aldehydes and ketones

- of the carbonyl compounds, aldehydes and ketones are the **simplest and the most important**
- **two systems of nomenclature** serve for them

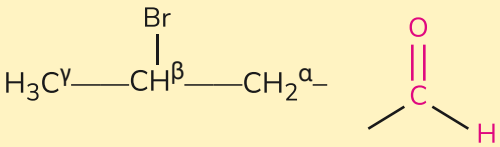
(a) Common names

- an aldehyde or a ketone is **more often called by its common name** than by its IUPAC one
- most aldehydes take their common name from the matching **carboxylic acid** (section B.1), with the acid's ending **–ic** replaced by **aldehyde**
- those names also carry the **Latin or Greek term for where the acid or aldehyde was first found**
- a substituent's place along the chain is marked by a **Greek letter** — α, β, γ, δ and so on
- the **α-carbon is the one bonded straight to the aldehyde group**, the **β-carbon the next**, and so on

Acetaldehyde

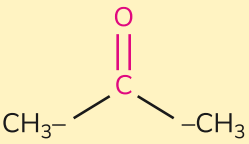


Benzaldehyde

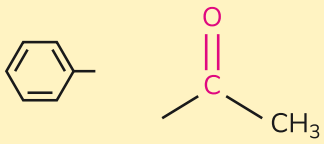


β-Bromobutyraldehyde

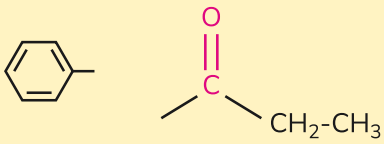
- name the **two alkyl or aryl groups** on the carbonyl group and a ketone's **common name** follows
- substituent positions carry **Greek letters — α α', β β'** and so on, counted outwards from the two carbons flanking the carbonyl group, which are α and α' themselves
- a few ketones hold **names inherited from history**: dimethyl ketone, the simplest, goes by **acetone**
- an **alkyl phenyl ketone** is usually named by putting the **acyl group's name before the word phenone**



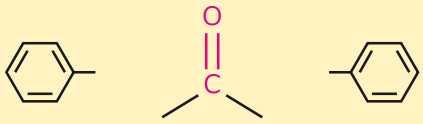
Acetone



Acetophenone



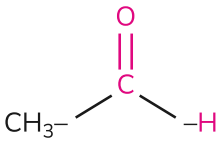
Propiophenone



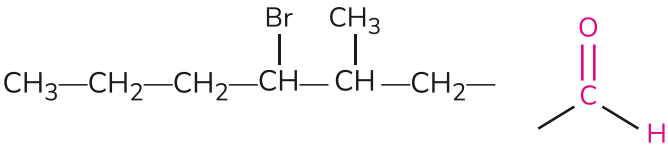
Benzophenone

(b) IUPAC names

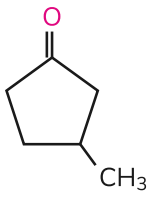
- an open chain aliphatic aldehyde or ketone takes its IUPAC name from the **matching alkane**, with the ending **–e replaced by –al** for an aldehyde and by **–one** for a ketone
- number the **longest carbon chain**: for an **aldehyde** begin at the **carbon of the aldehyde group**; for a **ketone** begin at **whichever end lies nearer the carbonyl group**
- substituents go in front, **in alphabetical order**, each with the numeral of its place on the chain
- a **cyclic ketone** follows the same rule, its **carbonyl carbon numbered one**
- where the **aldehyde group hangs on a ring**, add the suffix **carbaldehyde** after the full name of the cycloalkane, and **number the ring from the carbon the aldehyde group is attached to**
- the simplest **aromatic aldehyde**, with the aldehyde group on a benzene ring, is **benzenecarbaldehyde**; IUPAC accepts the common name **benzaldehyde** as well, so other aromatic aldehydes are named as **substituted benzaldehydes**



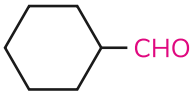
Ethanal



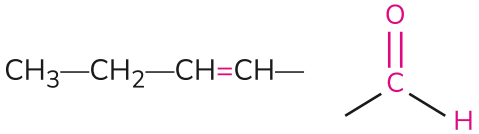
4-Bromo-3-methylheptanal



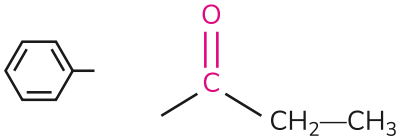
3-Methylcyclopentanone



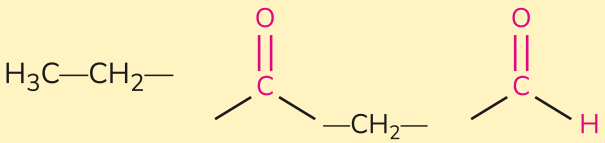
Cyclohexanecarbaldehyde



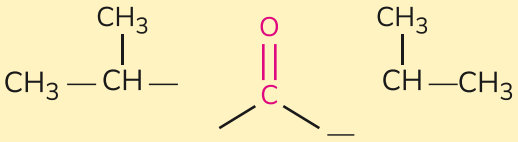
Pent-2-enal



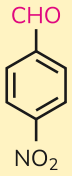
1-Phenylpropan-1-one



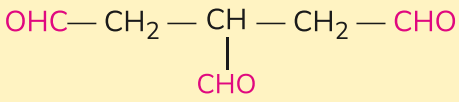
3-Oxopentanal



2,4-Dimethylpentan-3-one



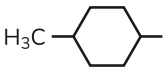
4-Nitrobenzenecarbaldehyde or 4-Nitrobenzaldehyde

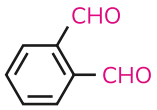
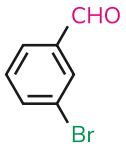
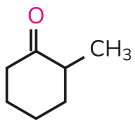


Propane-1,2,3-tricarbaldehyde

- naming it that way gives **all three aldehydic groups the same treatment**

TABLE 8.A **Common and IUPAC Names of Some Aldehydes and Ketones**

Structure	Common name · IUPAC name
Aldehydes	
$HCHO$	Formaldehyde · Methanal
CH_3CHO	Acetaldehyde · Ethanal
$(CH_3)_2CHCHO$	Isobutyraldehyde · 2-Methylpropanal
	γ-Methylcyclohexanecarbaldehyde 3-Methylcyclohexanecarbaldehyde
$CH_3CH(OCH_3)CHO$	α-Methoxypropionaldehyde · 2-Methoxypropanal
$CH_3CH_2CH_2CH_2CHO$	Valeraldehyde · Pentanal
$CH_2{=}CHCHO$	Acrolein · Prop-2-enal

Structure	Common name · IUPAC name
	Phthaldehyde Benzene-1,2-dicarbaldehyde
	*m*-Bromobenzaldehyde 3-Bromobenzenecarbaldehyde *or* 3-Bromobenzaldehyde
Ketones	
$CH_3COCH_2CH_2CH_3$	Methyl *n*-propyl ketone · Pentan-2-one
$(CH_3)_2CHCOCH(CH_3)_2$	Diisopropyl ketone · 2,4-Dimethylpentan-3-one
	α-Methylcyclohexanone 2-Methylcyclohexanone
$(CH_3)_2C{=}CHCOCH_3$	Mesityl oxide · 4-Methylpent-3-en-2-one

Structure of the carbonyl group

- the **carbonyl carbon is sp^2-hybridised** and forms **three sigma (σ) bonds**
- its **fourth valence electron stays in a *p*-orbital** and builds a **π-bond with oxygen**, by overlap with a *p*-orbital of that oxygen
- the oxygen also carries **two non bonding electron pairs**
- one plane therefore holds the **carbonyl carbon together with its three attached atoms**, and the **π-electron cloud lies above and below** it
- the bond angles come to roughly **120°**, as a trigonal coplanar structure requires (Figure 8.A)

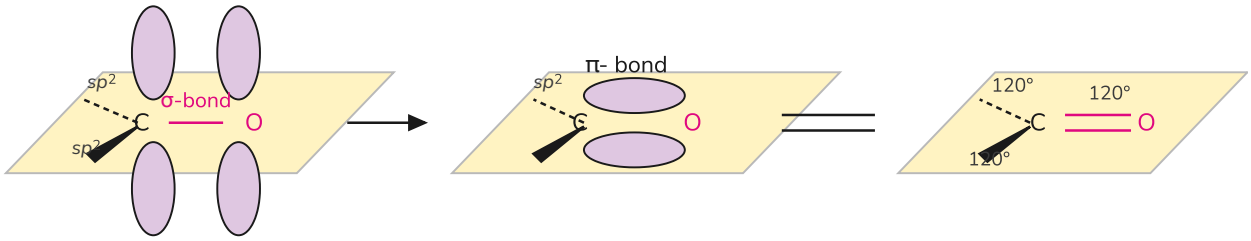
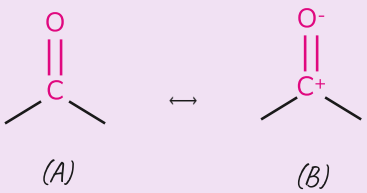


Figure 8.A Orbital diagram for the formation of carbonyl group



- the **carbon-oxygen double bond is polarised**, oxygen being the more electronegative
- so the **carbonyl carbon is an electrophilic (Lewis acid) centre** and the **carbonyl oxygen a nucleophilic (Lewis base) one**
- carbonyl compounds carry **substantial dipole moments** and are **more polar than ethers**
- the group's **high polarity** follows from resonance between a **neutral structure (A)** and a **dipolar one (B)**

Intext question

Q1 Draw the structure of each compound: (i) α-Methoxypropionaldehyde; (ii) 3-Hydroxybutanal; (iii) 2-Hydroxycyclopentane carbaldehyde; (iv) 4-Oxopentanal; (v) Di-sec. butyl ketone; (vi) 4-Fluoroacetophenone

A.2 Preparation of aldehydes and ketones

1 · By oxidation of alcohols

- oxidising a **primary alcohol** is the general route to an aldehyde, oxidising a **secondary** one the route to a ketone (Unit 7, Class XII)

2 · By dehydrogenation of alcohols

- this route suits **volatile alcohols** and is of **industrial application**
- the vapour of the alcohol is **passed over a heavy metal catalyst**, Ag or Cu
- an aldehyde comes off a **primary alcohol**, a ketone off a **secondary** one (Unit 7, Class XII)

3 · Starting from a hydrocarbon

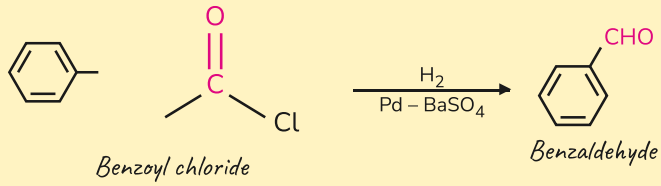
- **(i) By ozonolysis of alkenes** — ozonolyse the alkene, then treat with **zinc dust and water**, and aldehydes result, as sheet 6 sets out

- the products may be **ketones, or a mixture of both**, according to how the alkene is substituted (Unit 9, Class XI)
- **(ii) By hydration of alkynes** — water added to **ethyne over H_2SO_4 and $HgSO_4$** gives **acetaldehyde**; every other alkyne gives a **ketone** in this reaction (Unit 9, Class XI)

Preparation of aldehydes

1 · From acyl chloride (acid chloride)

- an acyl chloride is **hydrogenated over a catalyst, palladium on barium sulphate**; this is the **Rosenmund reduction**



Benzoyl chloride → Benzaldehyde

2 · From nitriles and esters

- a nitrile is **reduced to the matching imine by stannous chloride in hydrochloric acid**, and that imine on **hydrolysis gives the aldehyde**

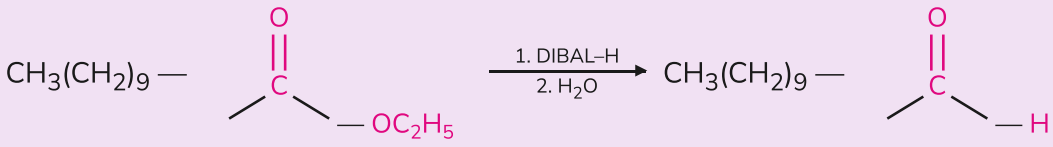
$$RCN + SnCl_2 + HCl \longrightarrow RCH = NH \xrightarrow{H_3O^+} RCHO$$

- this is the **Stephen** reaction
- nitriles may instead be **selectively reduced to imines by diisobutylaluminium hydride (DIBAL-H)**, then **hydrolysed to aldehydes**

$$RCN \xrightarrow[2.\ H_2O]{1.\ AlH(i\text{–}Bu)_2} R\text{–}CHO$$

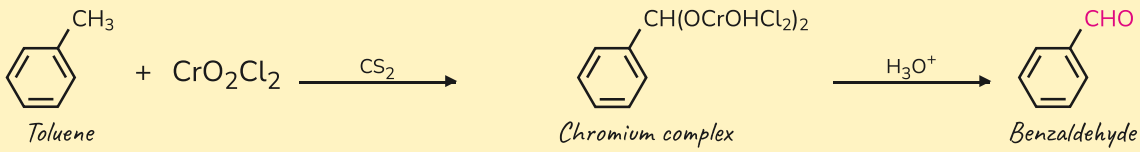
$$CH_3 — CH{=}CH\text{–}CH_2CH_2\text{–}CN \xrightarrow[2.\ H_2O]{1.\ AlH(i\text{–}Bu)_2} CH_3 — CH{=}CH\text{–}CH_2CH_2\text{–}CHO$$

- **esters too are reduced to aldehydes with DIBAL-H**

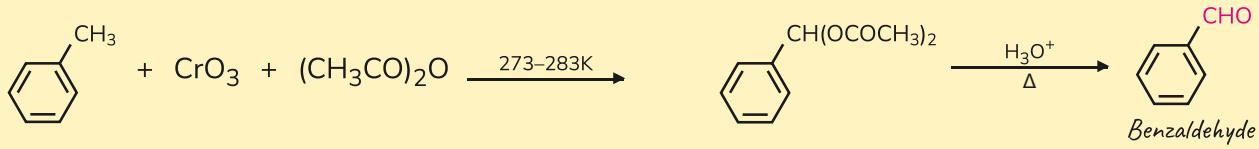


3 · From hydrocarbons

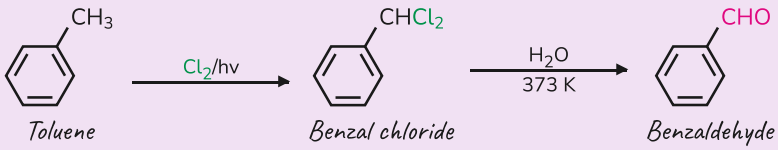
- benzaldehyde and its derivatives, the aromatic aldehydes, come from **aromatic hydrocarbons** by the methods below
- **(i) By oxidation of methylbenzene** — a **strong oxidising agent takes toluene and its derivatives all the way to benzoic acids**
- the oxidation can nevertheless be **held at the aldehyde stage**, by reagents that turn the **methyl group into an intermediate resistant to further oxidation**
- **(a) Use of chromyl chloride (CrO_2Cl_2)** — the methyl group is oxidised to a **chromium complex**, and **hydrolysing that complex** yields the matching benzaldehyde



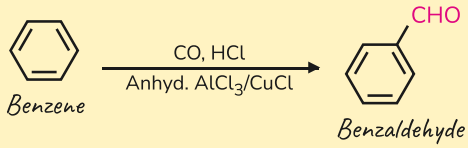
- this is the **Etard reaction**
- **(b) Use of chromic oxide (CrO_3)** — treat toluene, or a substituted one, with **chromic oxide in acetic anhydride** and **benzylidene diacetate** results
- **aqueous acid** then hydrolyses that diacetate to the matching **benzaldehyde**



- **(ii) By side chain chlorination followed by hydrolysis** — chlorinate the **side chain of toluene** to get **benzal chloride**, and **hydrolyse that** to benzaldehyde
- this is the **commercial method of manufacturing benzaldehyde**



- **(iii) By Gatterman – Koch reaction** — treat benzene or a derivative with **carbon monoxide and hydrogen chloride** over **anhydrous aluminium chloride or cuprous chloride**, and **benzaldehyde or a substituted benzaldehyde** results



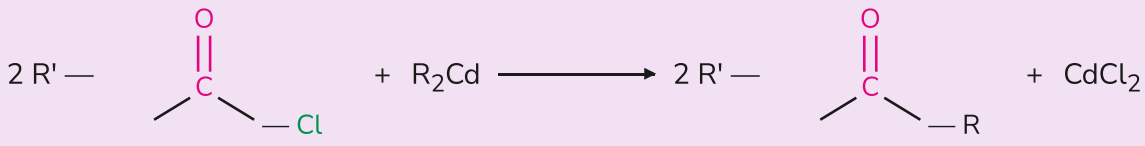
- this is the **Gatterman-Koch** reaction

Preparation of ketones

1 · From acyl chlorides

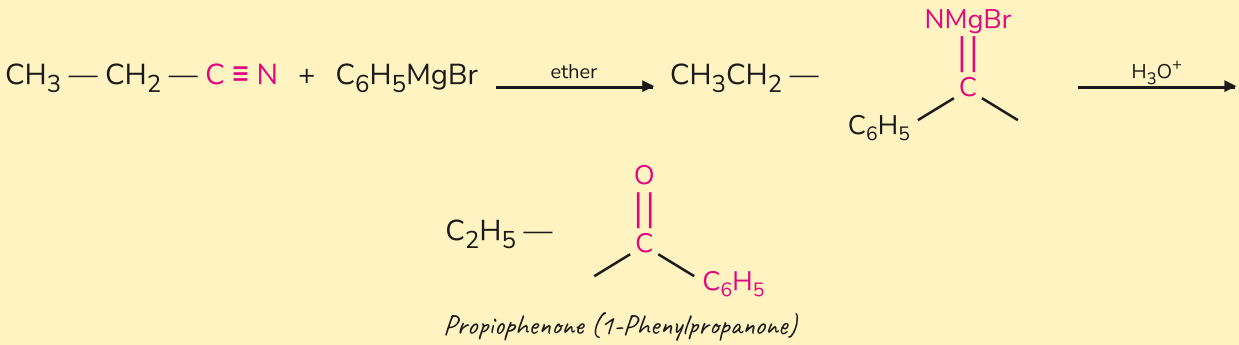
- acyl chlorides treated with **dialkylcadmium** give **ketones**
- the dialkylcadmium itself comes from **cadmium chloride with a Grignard reagent**





2 · From nitriles

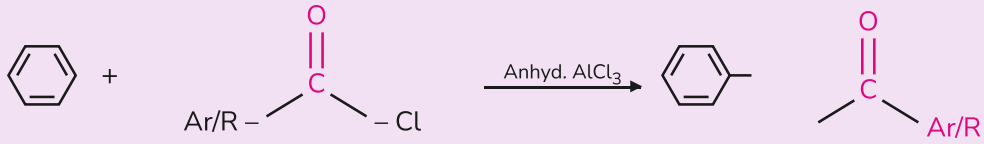
- a nitrile treated with a **Grignard reagent** and then **hydrolysed** yields a **ketone**



Propiophenone (1-Phenylpropanone)

3 · From benzene or substituted benzenes

- benzene, or a substituted benzene, treated with an **acid chloride over anhydrous aluminium chloride** affords the matching **ketone**
- this is the **Friedel-Crafts acylation reaction**



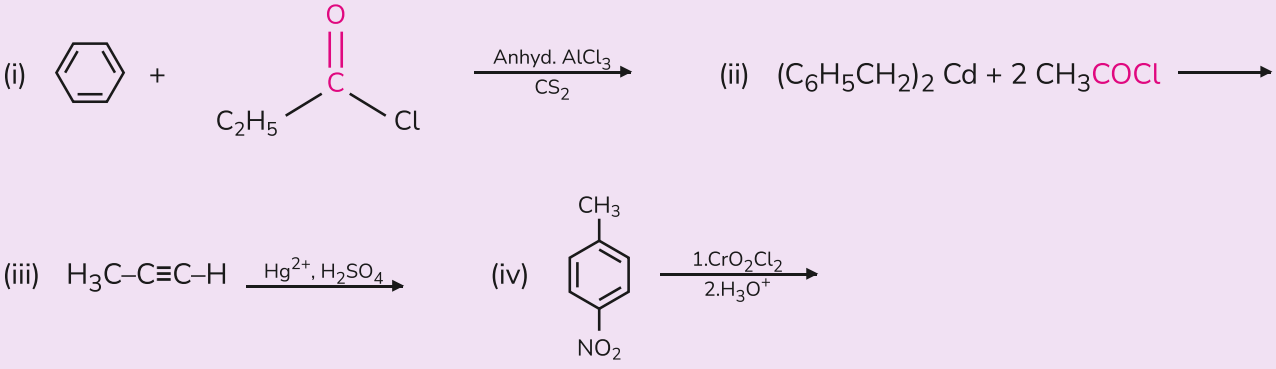
Worked example 8.a

Name the reagent that brings about each transformation.

- (i) Hexan-1-ol to hexanal · **$C_5H_5NH^+CrO_3Cl^-$ (PCC)**
- (ii) Cyclohexanol to cyclohexanone · **Anhydrous CrO_3**
- (iii) *p*-Fluorotoluene to *p*-fluorobenzaldehyde · **CrO_3 in acetic anhydride, or 1. CrO_2Cl_2 2. HOH**
- (iv) Ethanenitrile to ethanal · **(Diisobutyl)aluminium hydride (DIBAL-H)**
- (v) Allyl alcohol to propenal · **PCC**
- (vi) But-2-ene to ethanal · **O_3/H_2O-Zn dust**

Intext question

Q2 Draw the structures of the products of these reactions.

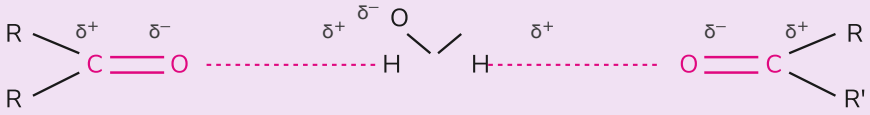


A.3 Physical properties

- **methanal is a gas** at room temperature, **ethanal a volatile liquid**; every other aldehyde and ketone is a **liquid or a solid** there
- they **boil above hydrocarbons and ethers of comparable molecular mass**, because **dipole-dipole interaction** associates their molecules weakly
- they **boil below alcohols of similar molecular mass**, having **no intermolecular hydrogen bonding**
- the compounds below, of molecular mass 58 and 60, are ranked by rising boiling point

	b.p. (K)	Molecular Mass
n-Butane	273	58
Methoxyethane	281	60
Propanal	322	58
Acetone	329	58
Propan-1-ol	370	60

- the **lower members — methanal, ethanal, propanone — mix with water in all proportions**, because they **hydrogen bond to it**



- solubility **falls away rapidly as the alkyl chain grows longer**
- all aldehydes and ketones dissolve fairly well in **organic solvents** — benzene, ether, methanol, chloroform and the like
- the **lower aldehydes smell sharp and pungent**; as the molecule grows the **odour turns less pungent and more fragrant**
- many naturally occurring aldehydes and ketones go into the **blending of perfumes and flavouring agents**

Worked example 8.b

Put these in order of rising boiling point: $CH_3CH_2CH_2CHO$, $CH_3CH_2CH_2CH_2OH$, $H_5C_2\text{-}O\text{-}C_2H_5$, $CH_3CH_2CH_2CH_3$.

- their molecular masses all lie between **72 and 74**
- only **butan-1-ol** molecules associate, and they do so by **extensive intermolecular hydrogen bonding**, so it **boils highest**
- **butanal is more polar than ethoxyethane**, so the **dipole-dipole attraction between its molecules is the stronger**
- between ***n*-pentane** molecules there is nothing but weak **van der Waals attraction**
- so the order runs: $CH_3CH_2CH_2CH_3 < H_5C_2\text{-}O\text{-}C_2H_5 < CH_3CH_2CH_2CHO < CH_3CH_2CH_2CH_2OH$

Intext question

Q3 Put these in order of rising boiling point: CH_3CHO, CH_3CH_2OH, CH_3OCH_3, $CH_3CH_2CH_3$.

A.4 Chemical reactions

- aldehydes and ketones **share the carbonyl functional group**, so they give **similar chemical reactions**

1 · Nucleophilic addition reactions

- alkenes take **electrophilic** addition; aldehydes and ketones take **nucleophilic** addition
- **(i) Mechanism** — the nucleophile attacks the **electrophilic carbon of the polar carbonyl group**, coming in **roughly perpendicular to the plane of the carbon's sp^2 orbitals** (Figure 8.B)
- the carbon's hybridisation **changes from sp^2 to sp^3**, and a **tetrahedral alkoxide intermediate** results
- the intermediate then **takes a proton out of the reaction medium**, which leaves the **product electrically neutral**
- net, **Nu^- and H^+ have been added across the C=O double bond**

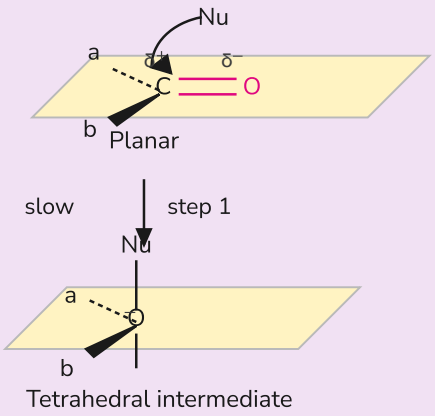


Figure 8.B Nucleophilic attack on carbonyl carbon

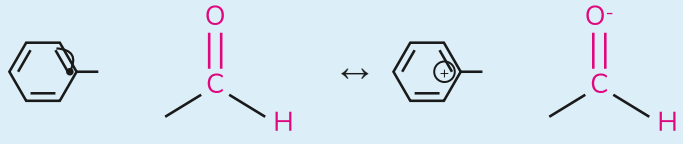
(ii) Reactivity

- in nucleophilic addition an **aldehyde generally outruns a ketone**, for reasons both **steric and electronic**
- **sterically**: a ketone carries **two relatively large substituents**, which hinder the nucleophile's approach to the carbonyl carbon more than the **one such substituent** of an aldehyde
- **electronically**: **two alkyl groups blunt the carbonyl carbon's electrophilicity more than one does**

Worked example 8.c

Would you expect benzaldehyde to be more or less reactive than propanal in nucleophilic addition? Explain.

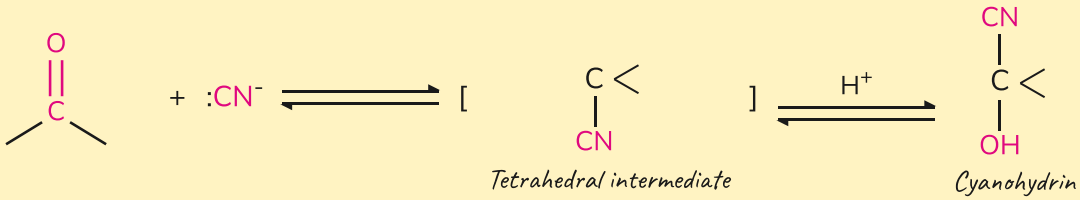
- the **carbonyl carbon of benzaldehyde is less electrophilic** than that of propanal
- **resonance cuts the polarity of the carbonyl group** in benzaldehyde, as below, so it is **less reactive than propanal**



(iii) The nucleophilic additions, and the addition-eliminations, that matter

- **(a) Addition of hydrogen cyanide (HCN)** — HCN set against an aldehyde or a ketone yields a **cyanohydrin**
- with **pure HCN the reaction is very slow**, so it is **catalysed by a base**: the **cyanide ion (CN^-) so generated is the stronger nucleophile** and adds readily to the carbonyl compound
- **cyanohydrins are useful synthetic intermediates**

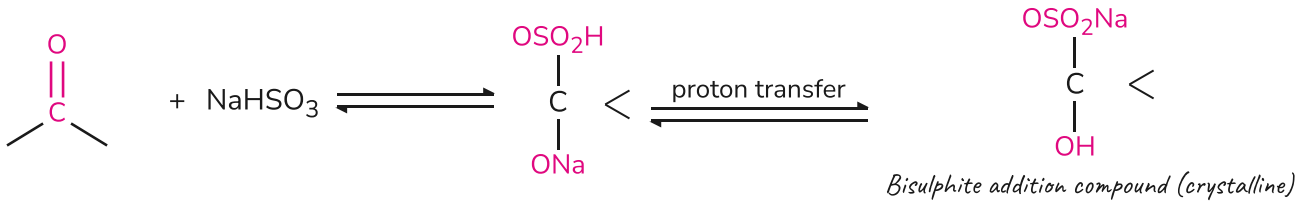
$$HCN + OH^- \rightleftharpoons :CN^- + H_2O$$



Tetrahedral intermediate — *Cyanohydrin*

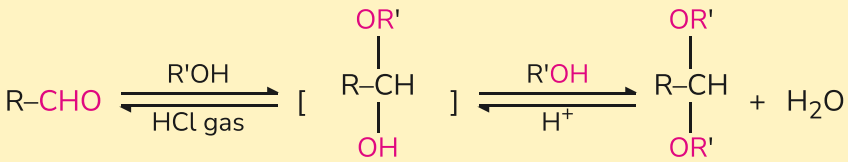
- **(b) Addition of sodium hydrogensulphite** — it adds across the carbonyl group to give the **addition products** below
- **steric reasons** put the equilibrium **well over to the right with most aldehydes**, and **over to the left with most ketones**
- the addition compound **dissolves in water**, and **dilute mineral acid or alkali restores the original carbonyl compound** from it
- that makes them useful for **separating and purifying aldehydes**

- **(c) Addition of Grignard reagents** — see Unit 7, Class XII

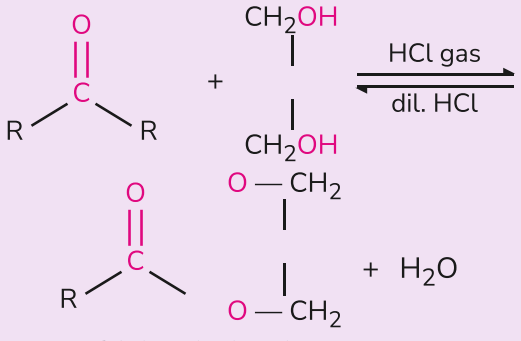


Bisulphite addition compound (crystalline)

- **(d) Addition of alcohols** — an aldehyde meets **one equivalent of a monohydric alcohol in dry hydrogen chloride** to yield an **alkoxyalcohol intermediate**, the **hemiacetal**
- that hemiacetal takes up **one more alcohol molecule** to give a ***gem*-dialkoxy compound, the acetal**



Hemiacetal — *Acetal*



Ethylene glycol ketal

- under similar conditions **ethylene glycol** turns a **ketone** into a **cyclic product, the ethylene glycol ketal**
- **dry hydrogen chloride puts a proton on the carbonyl oxygen**, and the **carbonyl carbon turns more electrophilic** for it — continued on sheet 12

- protonation thereby eases the **nucleophilic attack of ethylene glycol**
- **aqueous mineral acid hydrolyses acetals and ketals** back to the matching aldehydes and ketones

- **(e) Addition of ammonia and its derivatives — ammonia, and the H_2N-Z derivatives of it**, are nucleophiles that add across the carbonyl group
- acid catalyses the reaction, and it **runs both ways**
- the equilibrium **favours the product**, since the intermediate **dehydrates rapidly to >C=N-Z**
- Z = Alkyl, aryl, OH, NH_2, C_6H_5NH, $NHCONH_2$, and so on

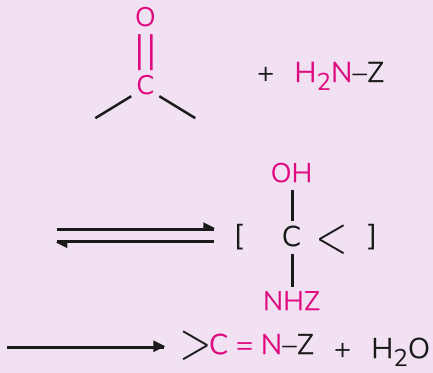


Table 8.B **Some N-Substituted Derivatives of Aldehydes and Ketones (>C=N-Z)**

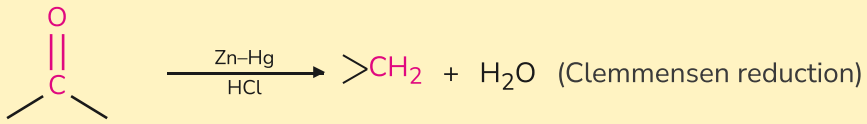
Z	Reagent name	Carbonyl derivative	Product name
-H	Ammonia	>C=NH	Imine
-R	Amine	>C=NR	Substituted imine (Schiff's base)
—OH	Hydroxylamine	>C=N–OH	Oxime
—NH_2	Hydrazine	>C=N–NH_2	Hydrazone
—HN–C$_6$H$_5$	Phenylhydrazine	>C=N-NH–C$_6$H$_5$	Phenylhydrazone
—HN–C$_6$H$_3$(2-O_2N, 4-NO_2)	2,4-Dinitrophenyl-hydrazine	>C=N-NH–C$_6$H$_3$(2-O_2N, 4-NO_2)	2,4 Dinitrophenyl-hydrazone
—NH—C(=O)—NH_2	Semicarbazide	>C=N-NH—C(=O)—NH_2	Semicarbazone

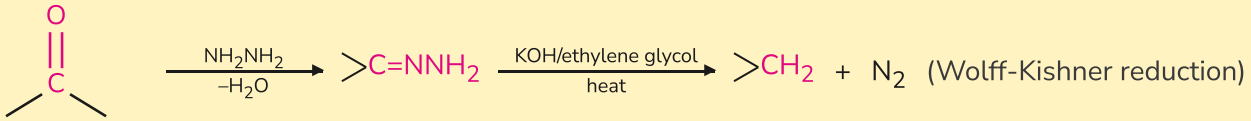
2,4-DNP-derivatives are yellow, orange or red solids, useful for characterisation of aldehydes and ketones.

2 · Reduction

- **(i) Reduction to alcohols — $NaBH_4$, $LiAlH_4$** or **catalytic hydrogenation** brings an aldehyde down to a **primary alcohol** and a ketone to a **secondary** one (Unit 7, Class XII)
- **(ii) Reduction to hydrocarbons — zinc-amalgam in concentrated hydrochloric acid** takes the carbonyl group down to a **CH_2 group**; this is the **Clemmensen** reduction — continued on sheet 13

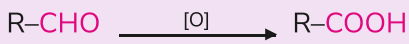
- the same reduction follows from **hydrazine, then heating with sodium or potassium hydroxide in a high boiling solvent** such as ethylene glycol — the **Wolff-Kishner reduction**





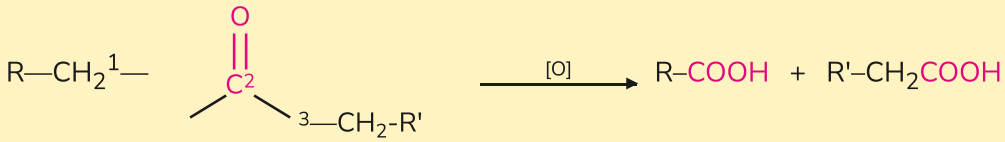
3 · Oxidation

- aldehydes and ketones **part company over oxidation**
- an **aldehyde is easily oxidised to a carboxylic acid** by common oxidising agents — nitric acid, potassium permanganate, potassium dichromate and the like
- even **mild oxidising agents**, chiefly **Tollens' reagent and Fehlings' reagent**, will oxidise an aldehyde



- Bernhard Tollens (1841-1918) held a Chair of Chemistry at Gottingen, in Germany

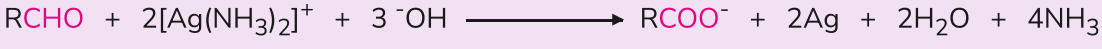
- ketones are oxidised only under **vigorous conditions**: a **strong oxidising agent at a raised temperature**
- their oxidation **cleaves a carbon-carbon bond**, affording a **mixture of carboxylic acids of fewer carbon atoms than the parent ketone**



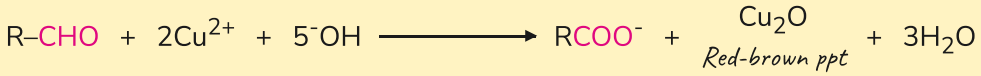
(by cleavage of the C_1-C_2 bond)

+ $R–CH_2COOH$ + $R'–COOH$ (by cleavage of the C_2-C_3 bond)

- the **mild oxidising agents below tell aldehydes from ketones**
- **(i) Tollens' test** — warm an aldehyde with Tollens' reagent, an **ammoniacal silver nitrate solution made up fresh**, and **silver metal** deposits as a **bright mirror**
- the aldehyde is oxidised to the matching **carboxylate anion**, the reaction running in **alkaline medium**

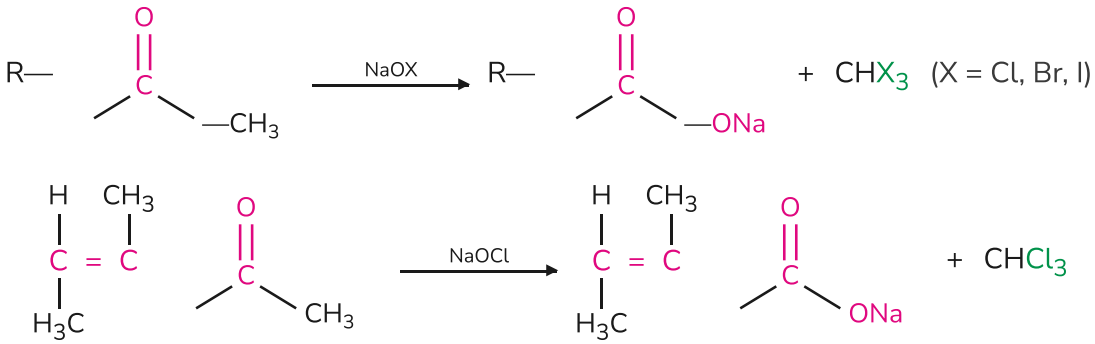


- **(ii) Fehling's test** — Fehling reagent is made of **two solutions**: **Fehling solution A, aqueous copper sulphate**, and **Fehling solution B, alkaline sodium potassium tartarate (Rochelle salt)**
- the two are **mixed in equal amounts** before the test
- heat an aldehyde with Fehling's reagent and a **reddish brown precipitate** comes down; the aldehyde is oxidised to the matching **carboxylate anion**
- an **aromatic aldehyde gives no response** to this test



- **(iii) Oxidation of methyl ketones by the haloform reaction** — an aldehyde or ketone with **at least one methyl group on the carbonyl carbon**, a **methyl ketone**, is oxidised by **sodium hypohalite**
- the products are the **sodium salt of the carboxylic acid one carbon shorter** than the carbonyl compound, and the **methyl group converted to haloform**
- this oxidation **leaves any carbon-carbon double bond in the molecule untouched**

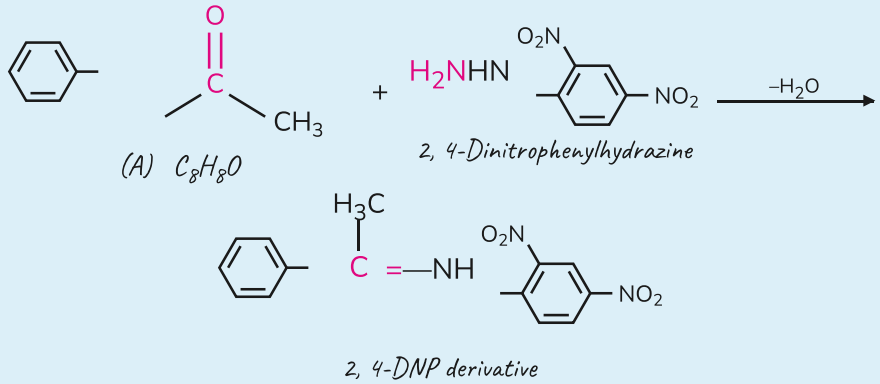
- the **iodoform reaction with sodium hypoiodite** also serves to detect a **CH_3CO group**, or a **$CH_3CH(OH)$ group**, which oxidation turns into one

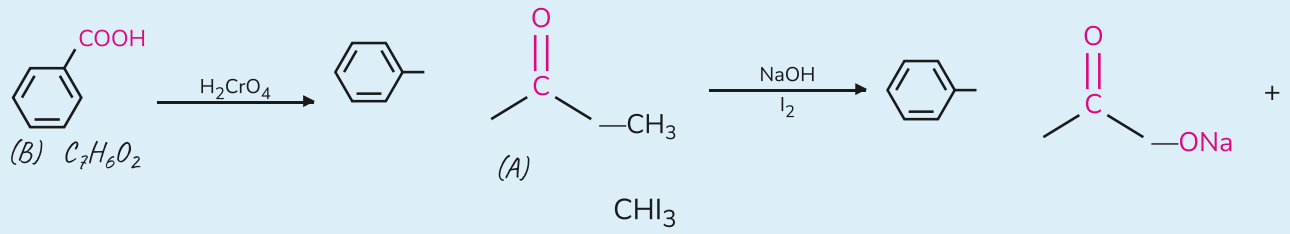


Worked example 8.d

An organic compound (A), molecular formula C_8H_8O, forms an orange-red precipitate with 2,4-DNP reagent and a yellow precipitate on heating with iodine in sodium hydroxide. It reduces neither Tollens' nor Fehlings' reagent, and it decolourises neither bromine water nor Baeyer's reagent. Drastic oxidation with chromic acid gives a carboxylic acid (B) of formula $C_7H_6O_2$. Identify (A) and (B) and explain the reactions.

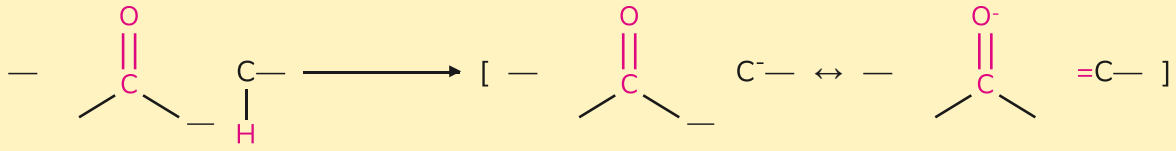
- (A) forms a **2,4-DNP derivative**, so it is an **aldehyde or a ketone**
- it **reduces neither Tollens' nor Fehling reagent**, so it must be a **ketone**
- it **responds to the iodoform test**, so it is a **methyl ketone**
- its molecular formula points to a **high degree of unsaturation**, yet it decolourises neither bromine water nor Baeyer's reagent: the unsaturation must belong to an **aromatic ring**
- (B), being the oxidation product of a ketone, should be a **carboxylic acid**; its formula makes it **benzoic acid**, so (A) is a **monosubstituted aromatic methyl ketone**
- reading (A)'s molecular formula, it is **phenyl methyl ketone (acetophenone)**



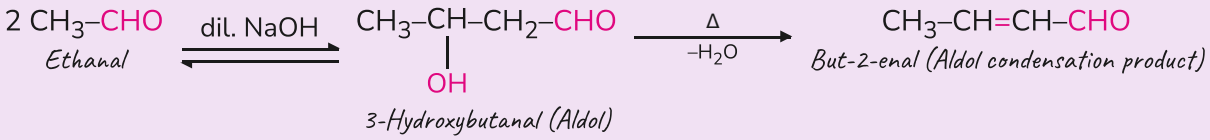


4 · Reactions due to α-hydrogen

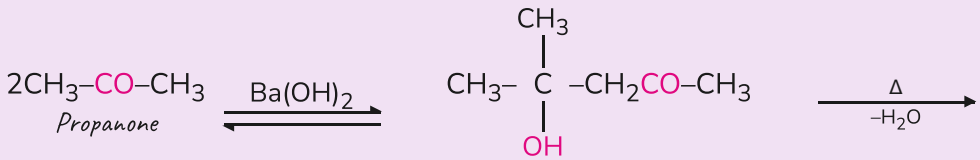
- aldehydes and ketones give a number of reactions that turn on the **acidic nature of their α-hydrogen**
- two things make it acidic: the **carbonyl group withdraws electrons strongly**, and **resonance stabilises the conjugate base**



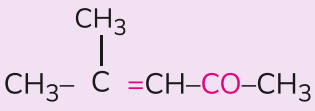
- **(i) Aldol condensation** — an aldehyde or ketone with **at least one α-hydrogen**, met by **dilute alkali as catalyst**, forms a **β-hydroxy aldehyde (aldol)** or a **β-hydroxy ketone (ketol)**
- this is the **Aldol reaction**



3-Hydroxybutanal (Aldol)



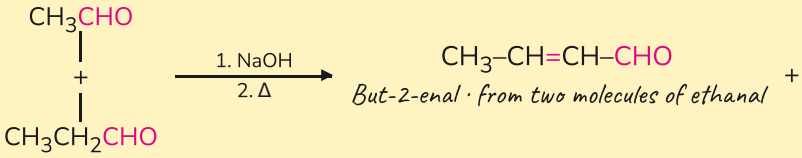
(Ketol) 4-Hydroxy-4-methylpentan-2-one

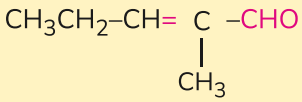


4-Methylpent-3-en-2-one (Aldol condensation product)

- the name **aldol** comes from the two functional groups the product carries, **aldehyde and alcohol**
- an aldol or a ketol **readily loses water** to give an **α,β-unsaturated carbonyl compound**, the **aldol condensation product**, and the reaction is called **Aldol condensation**
- ketones give **ketols**, which hold a **keto and an alcohol group**, yet the general name **aldol condensation still covers their reactions**, so like are they to the aldehydes'

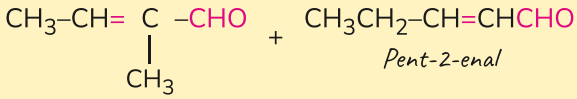
- **(ii) Cross aldol condensation** — run the aldol condensation between **two different aldehydes and/or ketones** and it is called **cross aldol condensation**
- where **both carry α-hydrogen atoms**, **four products** result, as the aldol reaction of a mixture of ethanal and propanal shows





2-Methylpent-2-enal · from two molecules of propanal

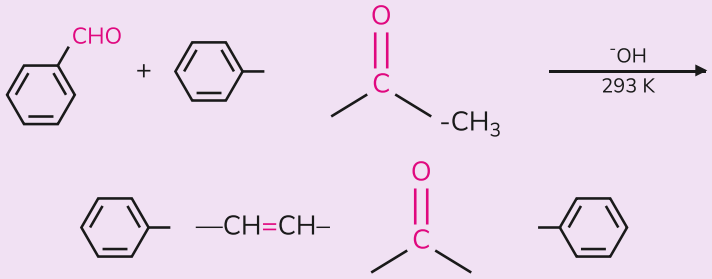
simple or self aldol products



2-Methylbut-2-enal *Pent-2-enal*

cross aldol products · from one molecule of ethanal and one of propanal

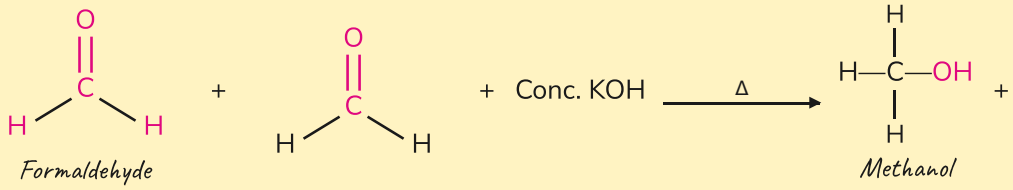
- **ketones too may serve as one component** of a cross aldol reaction



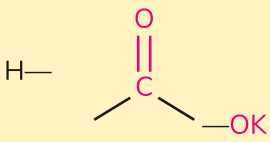
1, 3-Diphenylprop-2-en-1-one (Benzalacetophenone) (Major product)

5 · Other reactions

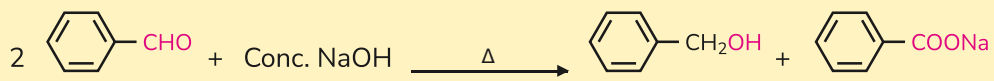
- **(i) Cannizzaro reaction** — an aldehyde with **no α-hydrogen atom** undergoes **self oxidation and reduction**, that is **disproportionation**, on heating with **concentrated alkali**
- one molecule of the aldehyde is **reduced to the alcohol**, another **oxidised to the salt of the carboxylic acid**



Formaldehyde *Methanol*

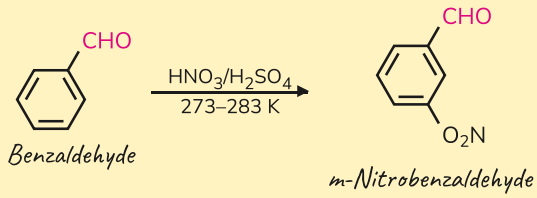


Potassium formate



Benzaldehyde *Benzyl alcohol* *Sodium benzoate*

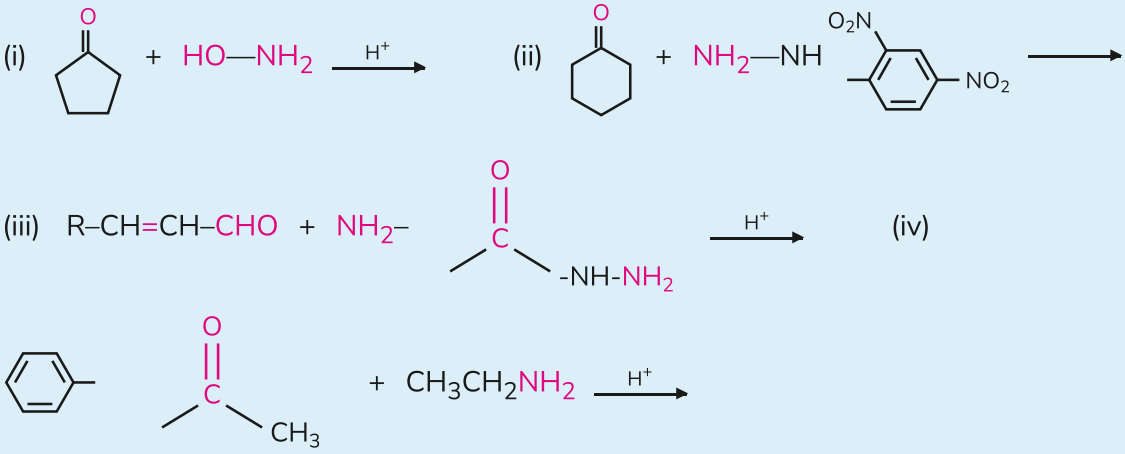
- **(ii) Electrophilic substitution reaction** — the ring of an aromatic aldehyde or ketone takes an electrophile, with the carbonyl group **deactivating it and directing to *meta***



Intext questions

Q4 Put each set in order of rising reactivity in nucleophilic addition. (i) Ethanal, propanal, propanone, butanone. (ii) Benzaldehyde, *p*-tolualdehyde, *p*-nitrobenzaldehyde, acetophenone. Hint: weigh the steric effect and the electronic effect.

Q5 Predict the products of these reactions.



A.5 Uses of aldehydes and ketones

- in the chemical industry aldehydes and ketones serve as **solvents, starting materials and reagents** for making other products
- **formaldehyde** is familiar as **formalin**, the 40% solution that **preserves biological specimens**; it also goes into **bakelite**, a phenol-formaldehyde resin, into **urea-formaldehyde glues** and into other polymers
- **acetaldehyde** serves chiefly as the starting material for **acetic acid and ethyl acetate, for vinyl acetate, for polymers and for drugs**
- **benzaldehyde** goes into **perfumery and the dye industries**
- **acetone** and **ethyl methyl ketone** are solvents in common industrial use
- many of them — **butyraldehyde, vanillin, acetophenone, camphor** — are known for their **odours and flavours**

II · Carboxylic acids

- carbon compounds holding the **carboxyl functional group, –COOH**, are the **carboxylic acids**
- the carboxyl group is a **carbonyl group with a hydroxyl group attached**, which is where its name comes from
- they are **aliphatic (RCOOH) or aromatic (ArCOOH)**, according to whether an **alkyl or an aryl group** sits on the carboxylic carbon
- **a great many carboxylic acids are found in nature**
- the **fatty acids**, that is the higher aliphatic members from **C_{12} to C_{18}**, are found in natural fats **esterified with glycerol**
- carboxylic acids are the **starting material for anhydrides, esters, acid chlorides and amides**

B.1 Nomenclature and structure of the carboxyl group

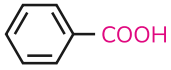
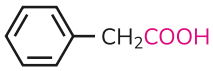
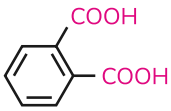
- carboxylic acids were among the **earliest organic compounds isolated from nature**, so a great many go by **common names**
- those names **end in the suffix –ic acid** and come from the **Latin or Greek name of the natural source**
- **formic acid (HCOOH)** first came from the **red ant** — Latin *formica*
- **acetic acid (CH_3COOH)** comes from **vinegar** — Latin *acetum*
- **butyric acid ($CH_3CH_2CH_2COOH$)** from **butter gone rancid** — Latin *butyrum*

- an aliphatic carboxylic acid takes its **IUPAC name** from the matching alkane, the ending **–e replaced by –oic acid**
- numbering the chain, the **carboxylic carbon is one**
- where **more than one carboxyl group** is present, number the **alkyl chain left by removing the carboxyl groups**, and add the multiplicative prefix ***dicarboxylic acid***, ***tricarboxylic acid*** and so on to the parent chain's name
- the **arabic numeral before the multiplicative prefix** gives the positions of the –COOH groups

TABLE 8.C Names and Structures of Some Carboxylic Acids

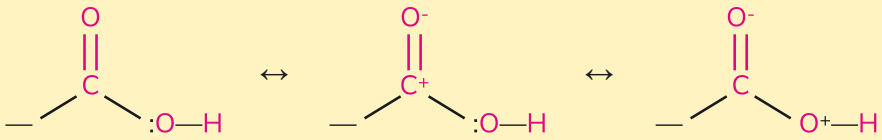
Structure	Common name · IUPAC name
$HCOOH$	Formic acid · Methanoic acid
CH_3COOH	Acetic acid · Ethanoic acid
CH_3CH_2COOH	Propionic acid · Propanoic acid
$CH_3CH_2CH_2COOH$	Butyric acid · Butanoic acid
$(CH_3)_2CHCOOH$	Isobutyric acid · 2-Methylpropanoic acid
$HOOC–COOH$	Oxalic acid · Ethanedioic acid

Structure	Common name · IUPAC name
$HOOC–CH_2–COOH$	Malonic acid · Propanedioic acid
$HOOC–(CH_2)_2–COOH$	Succinic acid · Butanedioic acid
$HOOC–(CH_2)_3–COOH$	Glutaric acid · Pentanedioic acid
$HOOC–(CH_2)_4–COOH$	Adipic acid · Hexanedioic acid
$HOOC–CH_2–CH(COOH)–CH_2–COOH$	Tricarballylic acid *or* carballylic acid Propane-1, 2, 3-tricarboxylic acid

Structure	Common name · IUPAC name
	Benzoic acid · Benzenecarboxylic acid (Benzoic acid)
	Phenylacetic acid · 2-Phenylethanoic acid
	Phthalic acid · Benzene-1, 2-dicarboxylic acid

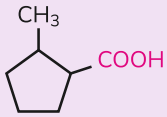
Structure of the carboxyl group

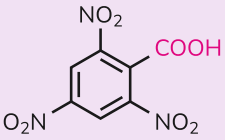
- one plane holds all the **bonds to the carboxyl carbon**, about **120°** apart
- the resonance structures below leave the **carboxylic carbon less electrophilic than a carbonyl carbon**



Intext question

Q6 Give each compound its IUPAC name.

(i) Ph CH_2CH_2COOH (ii) $(CH_3)_2C{=}CHCOOH$ (iii)  (iv)



B.2 Methods of preparing carboxylic acids

1 · From primary alcohols and aldehydes

- common oxidising agents take a **primary alcohol readily to the carboxylic acid**
- they are **$KMnO_4$**, in neutral, acidic or alkaline medium; **$K_2Cr_2O_7$**; and **CrO_3** under acid

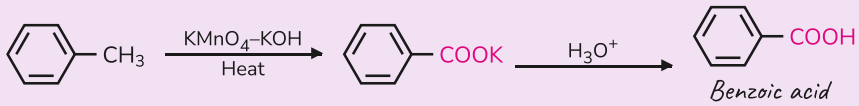
$$RCH_2OH \xrightarrow[2.\ H_3O^+]{1.\ \text{alkaline } KMnO_4} RCOOH$$

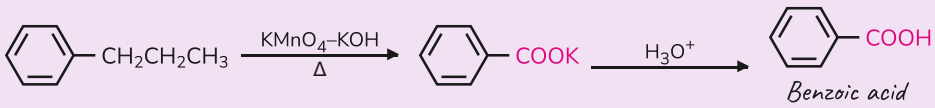
$$\underset{\textit{1-Decanol}}{CH_3(CH_2)_8CH_2OH} \xrightarrow[\text{Jones reagent}]{CrO_3\text{–}H_2SO_4} \underset{\textit{Decanoic acid}}{CH_3(CH_2)_8COOH}$$

- carboxylic acids also come **from aldehydes, with mild oxidising agents** (section A.4)

2 · From alkylbenzenes

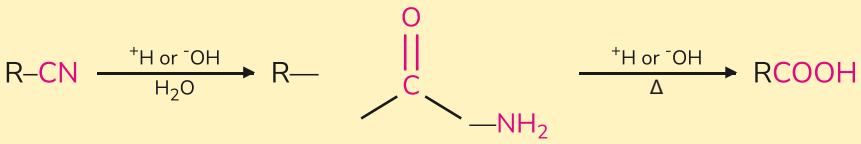
- oxidise an **alkyl benzene vigorously** — with **chromic acid**, or with **potassium permanganate under acid or alkali** — and an aromatic carboxylic acid results
- however long the side chain, **all of it goes to the carboxyl group**
- **primary and secondary alkyl groups** go this way; a **tertiary one is left alone**
- these reagents likewise carry **suitably substituted alkenes** to carboxylic acids

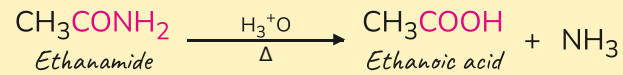


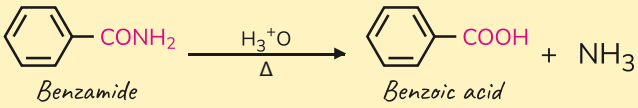


3 · From nitriles and amides

- a nitrile is **hydrolysed first to the amide and then to the acid**, over **H^+ or ^-OH as catalyst**
- **mild conditions stop the reaction at the amide stage**

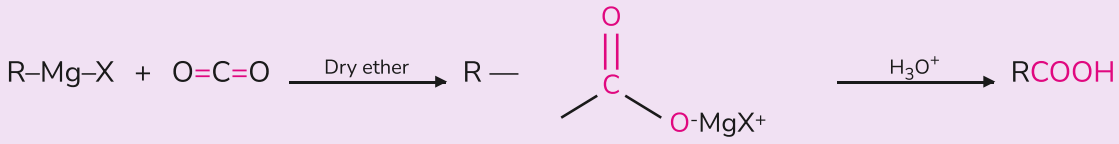






4 · From Grignard reagents

- a Grignard reagent meets **carbon dioxide (dry ice)** to form the **salt of a carboxylic acid**
- **acidifying that salt with mineral acid** gives the carboxylic acid itself



- Grignard reagents and nitriles are themselves prepared **from alkyl halides** (Unit 6, Class XII) — continued on sheet 21

- methods **3 and 4 turn alkyl halides into carboxylic acids of one carbon more** than the halide held — they **ascend the series**

5 · From acyl halides and anhydrides

- an **acid chloride hydrolysed with water** gives the carboxylic acid
- it hydrolyses **more readily with aqueous base**, to the **carboxylate ion**, and **acidification** then gives the acid
- **anhydrides**, for their part, are **hydrolysed to the matching acid or acids with water**

$$RCOCl \xrightarrow{H_2O} RCOOH + {}^-Cl$$

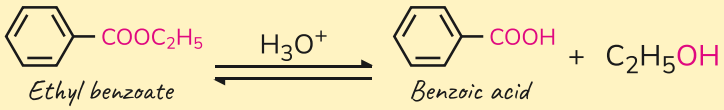
$$RCOCl \xrightarrow{{}^-OH/H_2O} RCOO^- + {}^-Cl \xrightarrow{H_3O^+} RCOOH$$

$$\underset{\text{Benzoic anhydride}}{(C_6H_5CO)_2O} \xrightarrow{H_2O} \underset{\text{Benzoic acid}}{2\ C_6H_5COOH}$$

$$\underset{\text{Benzoic ethanoic anhydride}}{C_6H_5COOCOCH_3} \xrightarrow{H_2O} \underset{\text{Benzoic acid}}{C_6H_5COOH} + \underset{\text{Ethanoic acid}}{CH_3COOH}$$

6 · From esters

- **acidic hydrolysis of an ester gives the carboxylic acid directly**
- **basic hydrolysis gives the carboxylate**, which on **acidification** gives the acid



$$\underset{\text{Ethyl butanoate}}{CH_3CH_2CH_2COOC_2H_5} \xrightleftharpoons{NaOH} CH_3CH_2CH_2COONa + C_2H_5OH \xrightarrow{H_3O^+} \underset{\text{Butanoic acid}}{CH_3CH_2CH_2COOH}$$

Worked example 8.e

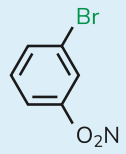
Write chemical reactions that bring about these transformations.

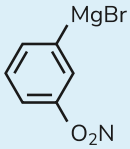
- (i) Butan-1-ol to butanoic acid
- (ii) Benzyl alcohol to phenylethanoic acid
- (iii) 3-Nitrobromobenzene to 3-nitrobenzoic acid
- (iv) 4-Methylacetophenone to benzene-1,4-dicarboxylic acid
- (v) Cyclohexene to hexane-1,6-dioic acid
- (vi) Butanal to butanoic acid

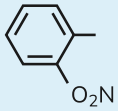
Worked example 8.e · the answer

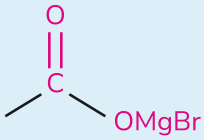
(i) $CH_3CH_2CH_2CH_2OH$ (*Butan-1-ol*) $\xrightarrow[\text{Jones reagent}]{CrO_3\text{–}H_2SO_4}$ $CH_3CH_2CH_2COOH$ (*Butanoic acid*)

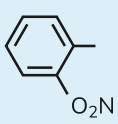
(ii) $C_6H_5CH_2OH$ (*Benzyl alcohol*) $\xrightarrow{HBr}$ $C_6H_5CH_2Br$ (*Benzyl bromide*) $\xrightarrow{KCN}$ $C_6H_5CH_2CN$ (*Benzyl cyanide*) $\xrightarrow[H_3O^+]{\Delta}$ $C_6H_5CH_2COOH$ (*Phenylethanoic acid*)

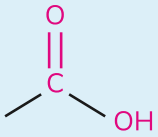
(iii) 

3-Nitrobromobenzene $\xrightarrow[\text{ether}]{Mg}$ 

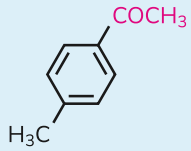
$\xrightarrow[\text{(dry ice)}]{CO_2}$ 

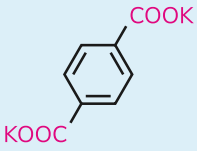


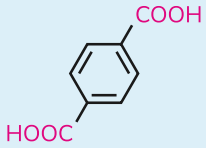
$\xrightarrow{H_3O^+}$ 



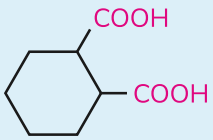
3-Nitrobenzoic acid

(iv) 

4-Methylacetophenone $\xrightarrow{KMnO_4/KOH}$ 

Dipotassium benzene-1, 4-dicarboxylate $\xrightarrow{dil.H_2SO_4}$ 

Benzene-1, 4-dicarboxylic acid (Terephthalic acid)

(v) 
Cyclohexene $\xrightarrow[\text{Heat}]{KMnO_4\text{–}H_2SO_4}$ 

Hexane-1, 6-dioic acid (Adipic acid)

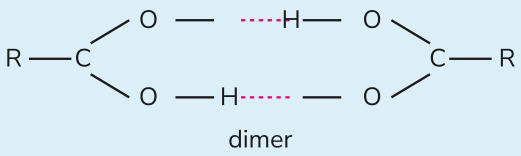
(vi) $CH_3CH_2CH_2CHO$ (*Butanal*) $\xrightarrow[\text{(Tollens' reagent)}]{\text{Ammoniacal } AgNO_3}$ $CH_3CH_2CH_2COOH$ (*Butanoic acid*)

Intext question

Q7 Show how each compound may be converted to benzoic acid: (i) Ethylbenzene; (ii) Acetophenone; (iii) Bromobenzene; (iv) Phenylethene (Styrene).

B.3 Physical properties

- up to **nine carbon atoms**, an aliphatic carboxylic acid is a **colourless liquid** at room temperature, and it **smells unpleasant**
- the **higher acids are solids, wax like**, and so **little volatile** as to be practically odourless
- they **boil above the aldehyde, the ketone and even the alcohol of like molecular mass, intermolecular hydrogen bonding associating their molecules more extensively**
- not even the **vapour phase breaks those hydrogen bonds completely**: most carboxylic acids stand as a **dimer there, and in an aprotic solvent too**



dimer

in the vapour state, or in an aprotic solvent

- simple aliphatic carboxylic acids of **up to four carbon atoms mix with water**, forming **hydrogen bonds with it**
- solubility **falls as the number of carbon atoms rises**: the **hydrocarbon part interacts more hydrophobically**
- water takes up **practically none of the higher carboxylic acids**
- **cold water barely dissolves benzoic acid**, the simplest aromatic carboxylic acid
- carboxylic acids also dissolve in **less polar organic solvents** — benzene, ether, alcohol, chloroform and the like

B.4 Chemical reactions

Reactions that cleave the O–H bond · acidity

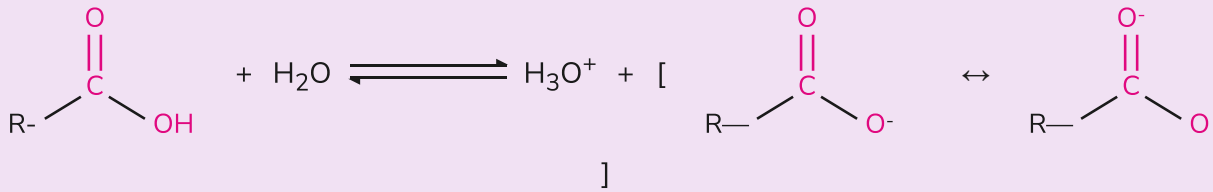
- as alcohols do, carboxylic acids **give off hydrogen with electropositive metals**; as phenols do, they **form salts with alkalies**
- unlike phenols, they react with **weaker bases — carbonates and hydrogencarbonates — to evolve carbon dioxide**
- that reaction is used to **detect a carboxyl group in an organic compound**

$$2R\text{–}COOH + 2Na \longrightarrow \underset{\text{Sodium carboxylate}}{2R\text{–}COO^-Na^+} + H_2$$

$$R\text{–}COOH + NaOH \longrightarrow R\text{–}COO^-Na^+ + H_2O$$

$$R\text{–}COOH + NaHCO_3 \longrightarrow R\text{–}COO^-Na^+ + H_2O + CO_2$$

- carboxylic acids **dissociate in water** to give a **resonance stabilised carboxylate anion** and a **hydronium ion**



- for that dissociation, K_{eq} is the **equilibrium constant** and K_a the **acid dissociation constant**

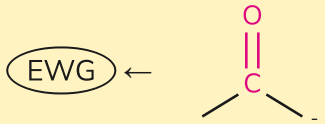
$$K_{eq} = \frac{[H_3O^+]\,[RCOO^-]}{[H_2O]\,[RCOOH]} \qquad K_a = K_{eq}\,[H_2O] = \frac{[H_3O^+]\,[RCOO^-]}{[RCOOH]}$$

- acid strength is usually quoted as a **pK_a value** rather than a K_a, where **pK_a = – log K_a**
- **hydrochloric acid** has pK_a **–7.0**; **trifluoroacetic acid**, strongest of the carboxylic acids, **0.23**; **benzoic acid 4.19**; **acetic acid 4.76**
- the **smaller the pK_a, the stronger the acid**, and the better it donates a proton
- **strong acids** sit below **pK_a 1**; those between **1 and 5** count as **moderately strong**; between **5 and 15** as **weak**; above **15** as **extremely weak**

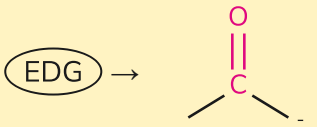
- set beside mineral acids they are **weak**, set beside alcohols and many simple phenols **strong** — pK_a runs about **16 for ethanol, 10 for phenol**
- they are in fact **among the most acidic organic compounds** met so far
- **two equivalent resonance structures** stabilise the **carboxylate ion**, the acid's conjugate base, and in each the negative charge rests on the **more electronegative oxygen**
- the conjugate base of a phenol, the **phenoxide ion**, has **non-equivalent resonance structures**, the negative charge sitting on the **less electronegative carbon**, so **resonance matters less there**
- a carboxylate also spreads its negative charge **over two electronegative oxygen atoms**, where a phenoxide spreads it **less effectively over one oxygen and several less electronegative carbons** (Unit 7, Class XII)
- so the **carboxylate ion is the better stabilised of the two**, and a **carboxylic acid the stronger acid**

Effect of substituents on the acidity

- a substituent may **alter the stability of the conjugate base**, and with it the acidity of the acid
- **electron withdrawing groups raise the acidity**, stabilising the conjugate base by **delocalising the negative charge through inductive and/or resonance effects**
- **electron donating groups lower the acidity**, destabilising the conjugate base



EWG props the carboxylate anion up; the acid strengthens



EDG unsettles the carboxylate anion; the acid weakens

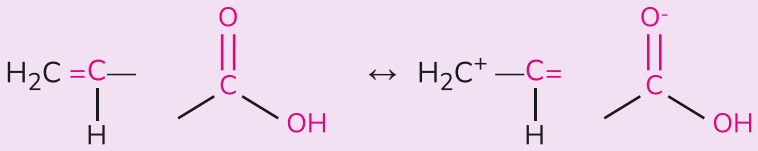
- these groups raise the acidity in the order **Ph < I < Br < Cl < F < CN < NO_2 < CF_3**
- so the acids below stand in order of **rising acidity**, read from right to left by their pK_a values

$CF_3COOH > CCl_3COOH > CHCl_2COOH > NO_2CH_2COOH > NC{-}CH_2COOH >$

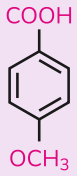
$FCH_2COOH > ClCH_2COOH > BrCH_2COOH > HCOOH > ClCH_2CH_2COOH >$

$C_6H_5COOH > C_6H_5CH_2COOH > CH_3COOH > CH_3CH_2COOH$

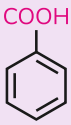
- attaching a **phenyl or a vinyl group straight to the carboxylic acid raises its acidity**, contrary to the fall the resonance effect below would lead one to expect



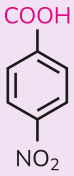
- the reason is the **greater electronegativity of the sp^2 hybridised carbon** the carboxyl carbon is attached to
- on an **aromatic** carboxylic acid, an **electron withdrawing group on the phenyl raises the acidity** and an **electron donating group lowers it**



4-Methoxy benzoic acid (pK_a = 4.46)



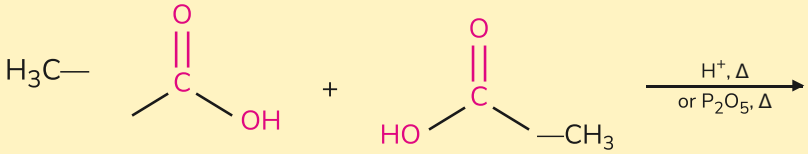
Benzoic acid (pK_a = 4.19)



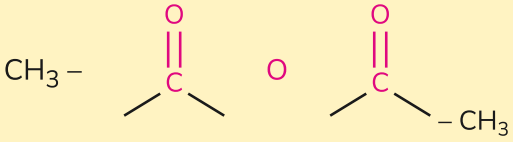
4-Nitrobenzoic acid (pK_a = 3.41)

Reactions that cleave the C–OH bond · 1 · Formation of anhydride

- carboxylic acids **heated with a mineral acid such as H_2SO_4, or with P_2O_5**, give the matching **anhydride**



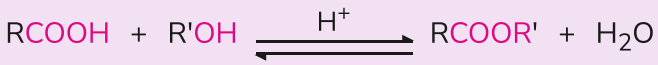
Ethanoic acid



Ethanoic anhydride

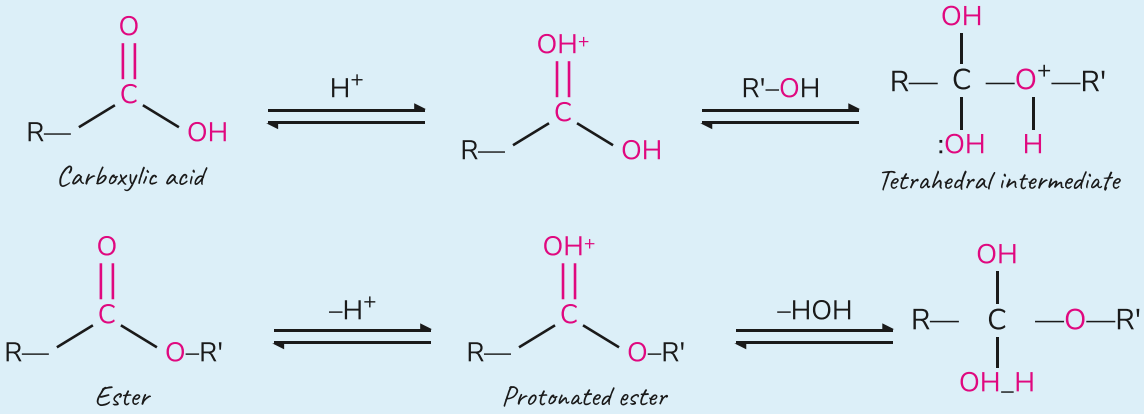
2 · Esterification

- an **alcohol or a phenol esterifies a carboxylic acid** over a **mineral acid catalyst**, concentrated H_2SO_4 or HCl gas



Mechanism of esterification of carboxylic acids

- esterifying a carboxylic acid with an alcohol is a kind of **nucleophilic acyl substitution**
- put a **proton on the carbonyl oxygen** and the **carbonyl group is activated** for the alcohol to add to it
- **proton transfer in the tetrahedral intermediate** turns the hydroxyl group into $-^{+}OH_2$, a **better leaving group**, which departs as a **neutral water molecule**
- the **protonated ester** left behind **gives up a proton**, and the ester remains



3 · Reactions with PCl_5, PCl_3 and $SOCl_2$

- a carboxylic acid's **hydroxyl group behaves as an alcohol's does**: treat with **PCl_5, PCl_3 or $SOCl_2$** and **chlorine readily takes its place**
- **thionyl chloride ($SOCl_2$) is preferred**: its other two products are **gases that escape**, which makes purifying the product easier

$$RCOOH + PCl_5 \longrightarrow RCOCl + POCl_3 + HCl$$

$$3RCOOH + PCl_3 \longrightarrow 3RCOCl + H_3PO_3$$

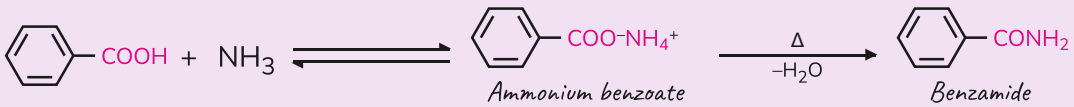
$$RCOOH + SOCl_2 \longrightarrow RCOCl + SO_2 + HCl$$

4 · Reaction with ammonia

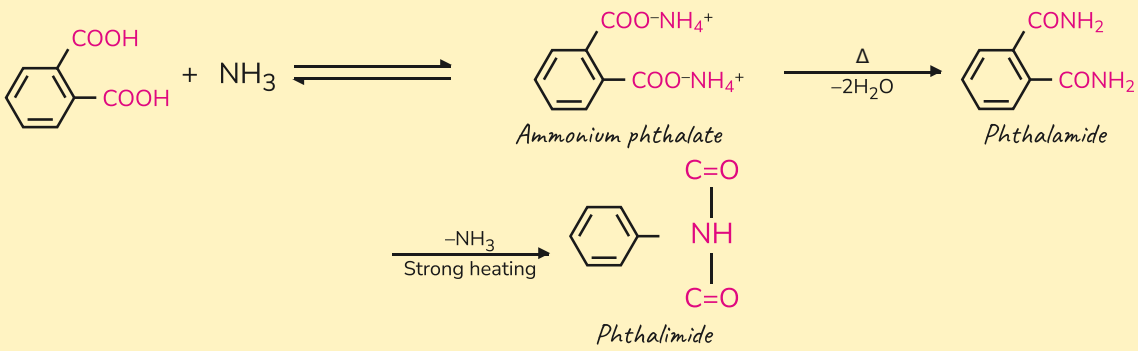
- ammonia gives the **ammonium salt**, and **heating that salt strongly** gives the amide

$$CH_3COOH + NH_3 \rightleftharpoons CH_3COO^{-}NH_4^{+} \xrightarrow[-H_2O]{\Delta} CH_3CONH_2$$

Ammonium acetate — Acetamide



Ammonium benzoate — Benzamide



Reactions involving the –COOH group · 1 · Reduction

- **lithium aluminium hydride reduces a carboxylic acid to the primary alcohol**, and **diborane** does it better
- ester, nitro, halo and like functional groups **resist diborane**
- the carboxyl group is **untouched by sodium borohydride**

$$R\text{–}COOH \xrightarrow[(ii)\ H_3O^+]{(i)\ LiAlH_4/ether\ or\ B_2H_6} R\text{–}CH_2OH$$

2 · Decarboxylation

- heat the **sodium salt with sodalime** — NaOH and CaO, 3 : 1 — and the acid **gives up carbon dioxide to leave a hydrocarbon**; that is **decarboxylation**

$$R\text{–}COONa \xrightarrow[Heat]{NaOH\ \&\ CaO} R\text{–}H + Na_2CO_3$$

- alkali metal salts of carboxylic acids also decarboxylate on **electrolysis of their aqueous solutions**, forming hydrocarbons with **twice the number of carbon atoms of the acid's alkyl group**
- this is **Kolbe electrolysis** (Unit 9, Class XI)

Substitution reactions in the hydrocarbon part · 1 · Halogenation

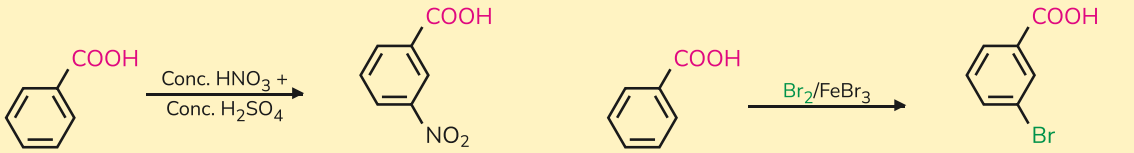
- a carboxylic acid with an **α-hydrogen** is **halogenated at the α-position** by **chlorine or bromine with a small amount of red phosphorus**, giving **α-halocarboxylic acids**
- this is the **Hell-Volhard-Zelinsky reaction**

$$R\text{–}CH_2\text{–}COOH \xrightarrow[(ii)\ H_2O]{(i)\ X_2/Red\ phosphorus} \underset{\alpha\text{ – Halocarboxylic acid}}{R\text{–}\underset{X}{CH}\text{–}COOH}$$

the α-carbon carries the halogen; X = Cl, Br

2 · Ring substitution

- aromatic carboxylic acids take **electrophilic substitution**, the carboxyl group acting as a **deactivating and meta-directing** group
- they do **not** undergo the **Friedel-Crafts reaction**: the carboxyl group deactivates the ring, and the catalyst **aluminium chloride (a Lewis acid) bonds to that carboxyl group**

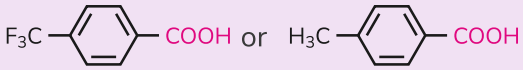


m-Nitrobenzoic acid *m-Bromobenzoic acid*

Intext question

Q8 Which acid of each pair below would you expect to be the stronger?

(i) CH_3CO_2H or CH_2FCO_2H (ii) CH_2FCO_2H or CH_2ClCO_2H (iii) $CH_2FCH_2CH_2CO_2H$ or $CH_3CHFCH_2CO_2H$

(iv) 

B.5 Uses of carboxylic acids

- **methanoic acid** serves **rubber and textiles, dyeing, leather, electroplating**
- **ethanoic acid** serves as a **solvent**, and as **vinegar in the food industry**
- **nylon-6, 6** is manufactured from **hexanedioic acid**
- **perfumery** uses the **esters of benzoic acid**, and **sodium benzoate preserves food**
- **higher fatty acids** go into **soaps and detergents**

Summary

- **aldehydes, ketones and carboxylic acids** are important classes of organic compound holding the **carbonyl group**, and their molecules are **highly polar**
- so they **boil above hydrocarbons, and above such weakly polar compounds** as the ethers of like molecular mass
- the **lower members dissolve in water**, **hydrogen bonding** to it; the **higher ones carry long hydrophobic carbon chains**, so **water will not take them, though common organic solvents will**
- aldehydes come from **primary alcohols dehydrogenated or oxidised under control**, and from **acyl halides reduced selectively**
- aromatic aldehydes also come from **methylbenzene oxidised by chromyl chloride, or by CrO_3 in acetic anhydride**; from **arenes formylated by carbon monoxide with hydrochloric acid, over anhydrous aluminium chloride and cuprous chloride**; and from **benzal chloride hydrolysed**
- ketones come from **secondary alcohols oxidised**, from **alkynes hydrated**, and from **acyl chloride set against dialkylcadmium**
- for an aromatic ketone the good method is **Friedel-Crafts acylation**, an aromatic hydrocarbon with an acyl chloride or an anhydride
- the **ozonolysis of alkenes** reaches either family
- they undergo **nucleophilic addition onto the carbonyl group** with nucleophiles such as **HCN, $NaHSO_3$, alcohols (or diols)** — continued on sheet 29

- they also add **ammonia derivatives** and **Grignard reagents**
- the **α-hydrogens of aldehydes and ketones are acidic**, so any of them carrying **at least one α-hydrogen undergoes Aldol condensation over a base**, giving **α-hydroxyaldehydes (aldol)** and **α-hydroxyketones (ketol)**
- lacking an α-hydrogen, an aldehyde takes the **Cannizzaro reaction under concentrated alkali** instead
- **$NaBH_4$, $LiAlH_4$ and catalytic hydrogenation** take them down to alcohols; the **Clemmensen and Wolff-Kishner reductions** take the carbonyl group all the way to a **methylene group**
- mild reagents such as **Tollens' and Fehling's oxidise an aldehyde readily to the carboxylic acid**, which is how **aldehyde is told from ketone**
- carboxylic acids come from **primary alcohols, aldehydes and alkenes oxidised**, from **nitriles hydrolysed**, and from **Grignard reagents met with carbon dioxide**; the aromatic ones also from **oxidising the side chain of an alkylbenzene**
- they are **a good deal more acidic than alcohols, and than most simple phenols**
- **$LiAlH_4$ reduces them to primary alcohols**, and **diborane in ether** does it better; **Cl_2 and Br_2 over red phosphorus halogenate them at the α-position**, in the Hell-Volhard Zelinsky reaction
- industry finds great use for **methanal and ethanal, propanone, benzaldehyde, formic acid, acetic acid and benzoic acid**

Exercises

E1 Explain each term below, and illustrate it with a reaction. (i) Cyanohydrin; (ii) Acetal; (iii) Semicarbazone; (iv) Aldol; (v) Hemiacetal; (vi) Oxime; (vii) Ketal; (viii) Imine; (ix) 2,4-DNP-derivative; (x) Schiff's base

E2 Give each compound its IUPAC name. (i) $CH_3CH(CH_3)CH_2CH_2CHO$; (ii) $CH_3CH_2COCH(C_2H_5)CH_2CH_2Cl$; (iii) $CH_3CH{=}CHCHO$; (iv) $CH_3COCH_2COCH_3$; (v) $CH_3CH(CH_3)CH_2C(CH_3)_2COCH_3$; (vi) $(CH_3)_3CCH_2COOH$; (vii) $OHCC_6H_4CHO$-*p*

E3 Draw the structures of these compounds. (i) 3-Methylbutanal; (ii) *p*-Nitropropiophenone; (iii) *p*-Methylbenzaldehyde; (iv) 4-Methylpent-3-en-2-one; (v) 4-Chloropentan-2-one; (vi) 3-Bromo-4-phenylpentanoic acid; (vii) *p,p′*-Dihydroxybenzophenone; (viii) Hex-2-en-4-ynoic acid

E4 Give the IUPAC names of these ketones and aldehydes, and the common names wherever there are any. (i) $CH_3CO(CH_2)_4CH_3$; (ii) $CH_3CH_2CHBrCH_2CH(CH_3)CHO$; (iii) $CH_3(CH_2)_5CHO$; (iv) Ph-CH=CH-CHO; (v) cyclohexanecarbaldehyde; (vi) PhCOPh

E5 Draw the structures of these derivatives. (i) The 2,4-dinitrophenylhydrazone of benzaldehyde; (ii) Cyclopropanone oxime; (iii) Acetaldehydedimethylacetal; (iv) The semicarbazone of cyclobutanone; (v) The ethylene ketal of hexan-3-one; (vi) The methyl hemiacetal of formaldehyde

E6 What does cyclohexanecarbaldehyde give with each of these? PhMgBr followed by H_3O^+; Tollens' reagent; semicarbazide in weak acid; ethanol in excess, with acid; zinc amalgam in dilute hydrochloric acid.

E7 Sort the compounds below into those that give an aldol condensation, those that give the Cannizzaro reaction, and those that give neither; then draw the products expected in each of the first two cases. (i) Methanal; (ii) 2-Methylpentanal; (iii) Benzaldehyde; (iv) Benzophenone; (v) Cyclohexanone; (vi) 1-Phenylpropanone; (vii) Phenylacetaldehyde; (viii) Butan-1-ol; (ix) 2,2-Dimethylbutanal

E8 Convert ethanal into each of butane-1,3-diol, but-2-enal and but-2-enoic acid.

E9 Propanal and butanal together can give four aldol condensation products. Draw each and name it, saying in every case which aldehyde is the nucleophile and which the electrophile.

E10 An organic compound of molecular formula $C_9H_{10}O$ forms a 2,4-DNP derivative, reduces Tollens' reagent and undergoes the Cannizzaro reaction. Vigorous oxidation gives 1,2-benzenedicarboxylic acid. Identify the compound.

E11 Dilute sulphuric acid hydrolysed an organic compound (A), $C_8H_{16}O_2$, to a carboxylic acid (B) and an alcohol (C). Chromic acid oxidised (C) to (B), and dehydrating (C) gave but-1-ene. Set out equations for every reaction.

E12 Put each set in order of the property named. (i) Acetaldehyde, acetone, di-*tert*-butyl ketone, methyl *tert*-butyl ketone — reactivity towards HCN. (ii) $CH_3CH_2CH(Br)COOH$, $CH_3CH(Br)CH_2COOH$, $(CH_3)_2CHCOOH$, $CH_3CH_2CH_2COOH$ — acid strength. (iii) Benzoic acid, 4-nitrobenzoic acid, 3,4-dinitrobenzoic acid, 4-methoxybenzoic acid — acid strength.

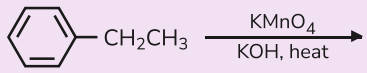
E13 Give a simple chemical test that tells each pair apart. (i) Propanal and propanone; (ii) Acetophenone and benzophenone; (iii) Phenol and benzoic acid; (iv) Benzoic acid and ethyl benzoate; (v) Pentan-2-one and pentan-3-one; (vi) Benzaldehyde and acetophenone; (vii) Ethanal and propanal

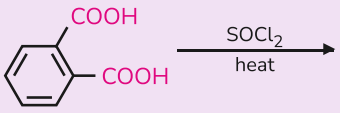
E14 Prepare each compound below starting from benzene. Any inorganic reagent may be used, and any organic reagent holding a single carbon atom. (i) Methyl benzoate; (ii) *m*-Nitrobenzoic acid; (iii) *p*-Nitrobenzoic acid; (iv) Phenylacetic acid; (v) *p*-Nitrobenzaldehyde

E15 Carry out each conversion below in two steps at most. (i) Propanone to propene; (ii) Benzoic acid to benzaldehyde; (iii) Ethanol to 3-hydroxybutanal; (iv) Benzene to *m*-nitroacetophenone; (v) Benzaldehyde to benzophenone; (vi) Bromobenzene to 1-phenylethanol; (vii) Benzaldehyde to 3-phenylpropan-1-ol; (viii) Benzaldehyde to α-hydroxyphenylacetic acid; (ix) Benzoic acid to *m*-nitrobenzyl alcohol

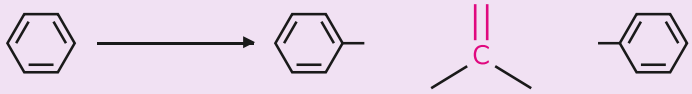
E16 Give an account of acetylation; of the Cannizzaro reaction; of cross aldol condensation; and of decarboxylation.

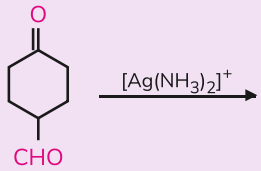
E17 Complete each synthesis by supplying the missing starting material, reagent or product.

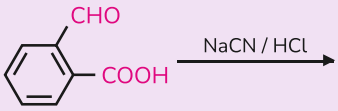
(i) 

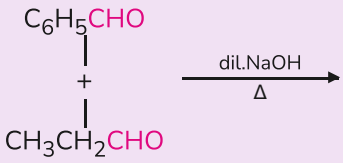
(ii) 

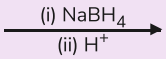
(iii) C_6H_5CHO 

(iv) 

(v) 

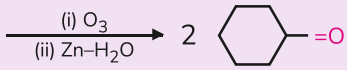
(vi) 

(vii) 

(viii) $CH_3COCH_2COOC_2H_5$ 

(ix) 

(x) 

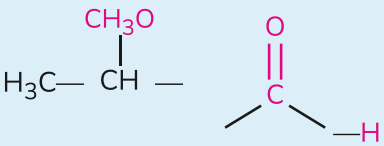
(xi) 

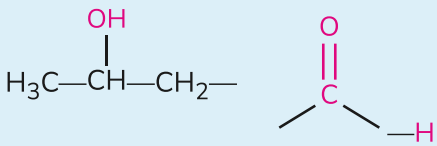
E18 Give a plausible explanation for each of these. (i) Cyclohexanone forms a cyanohydrin in good yield, but 2,2,6-trimethylcyclohexanone does not. (ii) Semicarbazide holds two $-NH_2$ groups, yet only one takes part in forming a semicarbazone. (iii) Preparing an ester from a carboxylic acid and an alcohol over an acid catalyst, the water or the ester must be removed as soon as it forms.

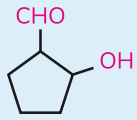
E19 An organic compound of molecular mass 86 holds 69.77% carbon and 11.63% hydrogen, the rest oxygen. Tollens' reagent it leaves alone, but sodium hydrogensulphite gives an addition compound with it and the iodoform test comes out positive. Oxidised vigorously it yields ethanoic and propanoic acid. Give a possible structure.

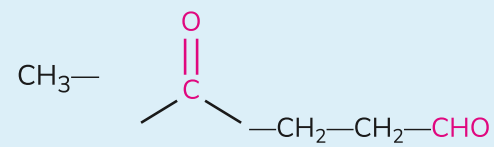
E20 More resonating structures belong to the phenoxide ion than to the carboxylate ion. Why, then, is the carboxylic acid the stronger acid?

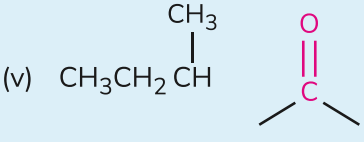
Answers · Q1

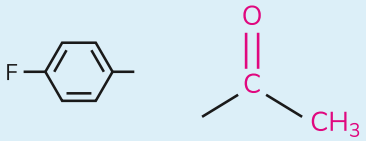
(i) 

(ii) 

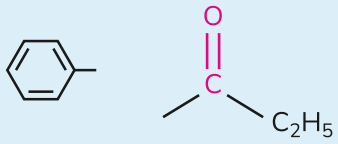
(iii) 

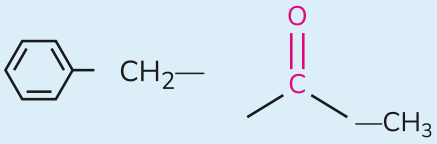
(iv) 

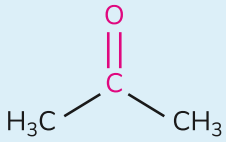
(v) 

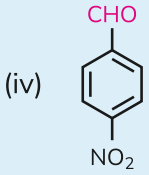
(vi) 

Q2

(i) 

(ii) 

(iii) 

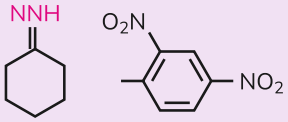
(iv) 

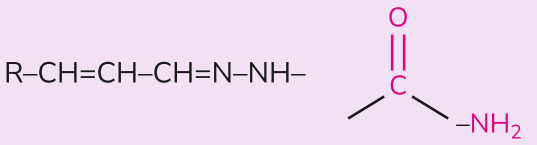
- **Q3** $CH_3CH_2CH_3 < CH_3OCH_3 < CH_3CHO < CH_3CH_2OH$
- **Q4** (i) Butanone < Propanone < Propanal < Ethanal
- (ii) Acetophenone < *p*-Tolualdehyde, Benzaldehyde < *p*-Nitrobenzaldehyde

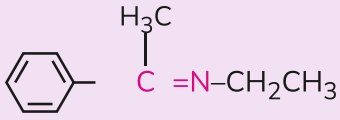
- **Q6** (i) 3-Phenylpropanoic acid; (ii) 3-Methylbut-2-enoic acid; (iii) 2-Methylcyclopentanecarboxylic acid; (iv) 2,4,6-Trinitrobenzoic acid
- **Q8** (i) CH_2FCOOH; (ii) CH_2FCOOH; (iii) CH_3CHFCH_2COOH; (iv) $F_3C–C_6H_4–COOH$

Q5

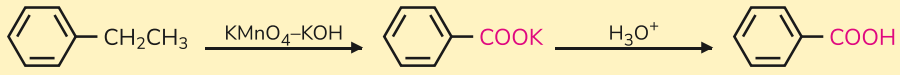
(i) 

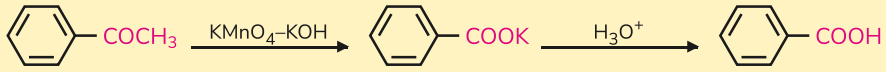
(ii) 

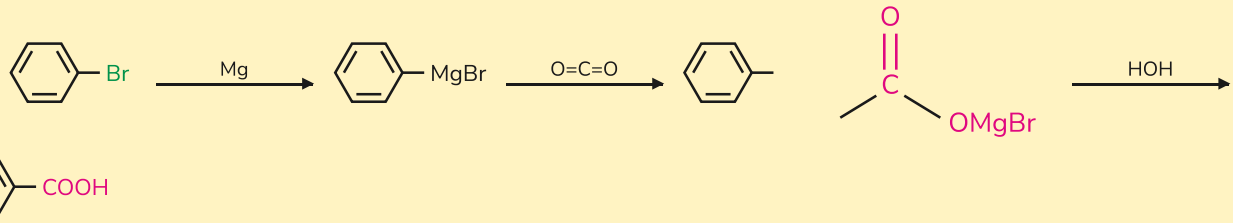
(iii) 

(iv) 

Q7

(i) 

(ii) 

(iii) 

(iv) 